

Krish Patel

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SUMMARY OF QUALIFICATIONS

- Strong foundation in mechanical design, electronics, and control systems integration.
- Proficient in MATLAB, Simulink, SolidWorks, and Arduino Programming with C/C++.
- Strong analytical and problem-solving abilities with a passion for Artificial Intelligence and intelligent system designs.

TECHNICAL SKILLS

- **Software & Analysis:** | **Open CV Sensor Integration** (Expert) | **SolidWorks** (Expert) | **C++ Programming** (Expert) | **Arduino Programming** (Expert) | **OnShape** (Expert) | **MATLAB** (Intermediate) | **Python Programming** (Expert).
- **Manufacturing:** 3-D Printing | CNC Machining | PCB Prototype and Circuit Design

EDUCATION

Bachelor of Engineering (BEng) – Mechatronics Engineering

Sep. 2023 - Expected Apr. 2028

Toronto Metropolitan University

Toronto, ON

- **Accomplishments:** | Winner of MET Junior Design Competition (November 2024) | Winner of OEC Junior Design Competition (January 2025) |
- **Relevant Coursework:** Programming, Analysis of Electric Circuits, Signals and Systems, Microprocessor Systems, Machine Mechanics and Design, Robotics, Computer Structures and Real-Time Systems, Electromechanical System Modelling and Design, Measurements, Sensors, and Instruments.

ENGINEERING PROJECTS

Smart Parking Garage System – Senior Design Project

Nov 2025 – Dec 2025

Metropolitan Engineering Competition (MEC) 2025

- Designed and built a multi-level smart parking garage prototype using Arduino-based control, integrating sensors, servo-driven gates, and real-time LED status indicators.
- Implemented AI-inspired parking allocation, safety systems, and energy management, including hazard detection, automated responses, and off-peak power optimization.

Robot Design and Artificial Intelligence Programming

Jan 2025 – May 2025

Autonomous Rover System Project (Academic Project)

- Designed and developed an autonomous rover system with object identification, obstacle avoidance, and self-navigation capabilities using computer vision (OpenCV, YOLOv4-tiny)
- Engineered optimized C++ architecture and hardware integration (Raspberry Pi Camera V2, GPU acceleration), improving real-time performance

DESIGN TEAMS

Research and Development Specialist

Aug 2025 - Present

Metropolitan Hyperloop

- Direct the design for the upcoming entry in the Aeroshell & Braking System category in the Hyperloop Global Competition.
- Redesign 35+ CAD components across 6 subsystems, focusing on Design for Manufacturing and Design for Assembly

Director of Outreach

June 2025 - Present

Mechatronics Course Union – Toronto Metropolitan University

- Partnered with faculty and industry experts to host skill-development workshops on topics like SolidWorks design, Arduino Programming, and embedded systems.
- Led industry and community outreach initiatives, building partnerships with engineering firms, start-ups, and university departments to create collaboration and sponsorship opportunities.

WORK EXPERIENCE

Lifeguard

May 2025 – Sept 2025

Markham Pan AM Centre

- Ensured patron safety by monitoring swimming areas and responding swiftly to emergencies with rescues, first aid, and CPR
- Collaborated with lifeguard teams to conduct emergency drills, improve response times, and educate swimmers on water safety and risk prevention.